

Agentic-J chose BoneJ and reproduced Table 2 of the paper:
3 of 4 metrics within 0.9%, Tb.Sp +5 to +10% from an unreported setting

Agentic-J chose Coloc 2 and reproduced the stress effect:
Δ mean R +0.50, published +0.49, 17 image fields

29 tracks over 101 frames, 1998 of 1998 spots on published cells;
mean speed 2.00 vs 2.02 µm/min published, r = 0.99

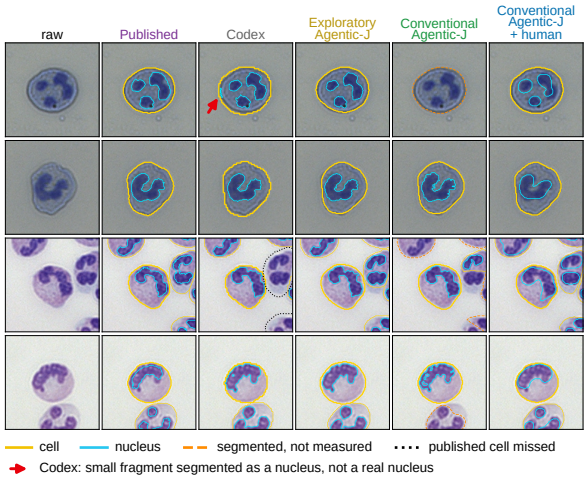
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Analysis	Time	Scientific verdict	Biologist feedback
CD177 neutrophil morphology			
Published	weeks	no meaningful difference	reference segmentation
Codex CLI	13 min	0/30 features significant	many cells missed; faulty cell and nucleus segmentation
Exploratory Agentic-J	35 min	5/25 significant, 3/3 runs	faulty nucleus segmentation; false-positive cells
Conventional Agentic-J	67 min	0/32 significant, 3/3 runs	cells segmented but discarded; nuclei missed or partly segmented
Conventional Agentic-J + human	488 min	0/11 tested significant 34 features extracted	clean cell segmentation; nuclei rounded and over-segmented

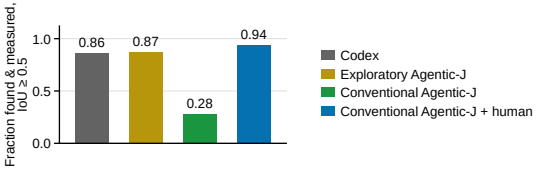
Time T1 / T2	Task 1 TUNEL+ and TUNEL− nuclei 6 nuclei, fixed frame	Task 2 proliferation in TUNEL+ hosts 15 parasites, last live frame
Leishmania infection: TUNEL and proliferation		
not recorded	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts 8 TUNEL+ vs 7 TUNEL−, p = 0.0012
18 min, both tasks	TUNEL+: nuclei 3, 6 TUNEL−: nuclei 1, 2, 4, 5	higher proliferation in TUNEL+ hosts p = 0.017; 10 of 15 parasites
22 / 69 min	TUNEL+: nucleus 3, split in two TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no parasite found 22 false detections, no host
55 / 85 min	TUNEL+: nucleus 3 TUNEL−: nuclei 1, 2, 4, 5, 6	not testable: no TUNEL+ host found 3 of 15 parasites, 2 false
34 / 26 min	TUNEL+: nucleus 3 TUNEL−: nuclei 1, 2, 4, 5, 6	higher proliferation in TUNEL+ hosts p = 0.0012; all 15 parasites, 14 within 0.1

Manual (analyst) · Human feedback (user) · Automatic (Agentic-J) · Automatic (other agent)

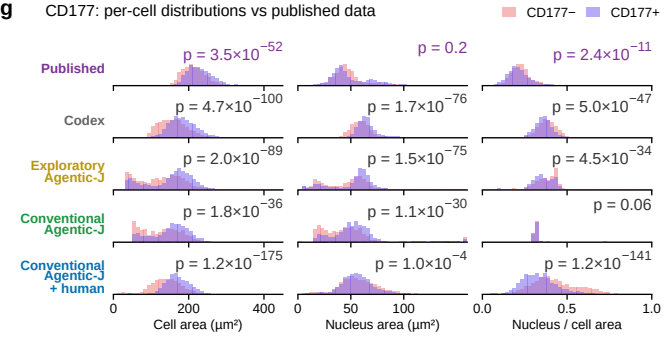
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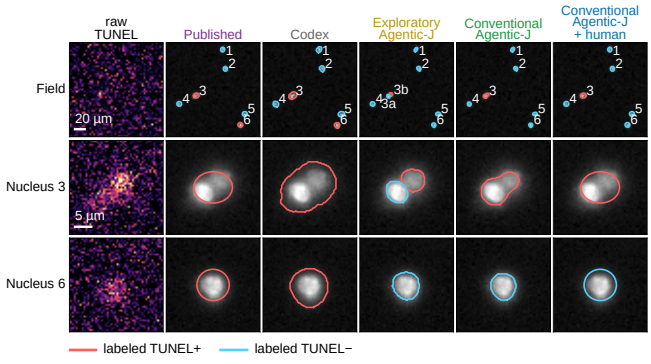
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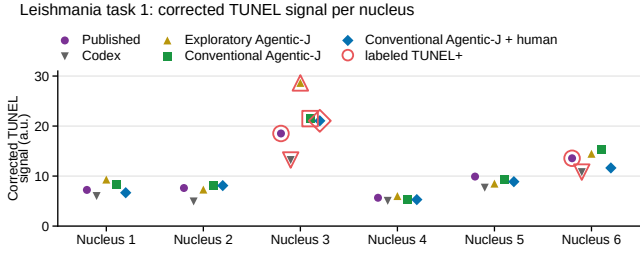
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